



ProcureFlow

AI-Powered Enterprise Procurement Platform

Cloud Native Microservices | GitOps | DevSecOps | Amazon EKS



Presented By
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Cloud DevOps Capstone Review | June 2026



THE CHALLENGE

The Problem We're Solving

Where Procurement Breaks Down Today



Fragmented systems

Vendors, purchase orders, invoices, contracts and documents scattered across spreadsheets and disconnected legacy tools.



No centralized visibility

No single source of truth for vendor risk, contract expiry timelines, or invoice anomalies across the business.



Manual, reactive processes

Risk and compliance issues are discovered after the fact, not flagged proactively before they cause loss.



Our Objective

Build a secure, scalable, cloud-native procurement platform with AI-assisted risk analysis — delivered using production-grade DevOps practices.

- Infrastructure as Code
- GitOps Delivery
- CI/CD Security Gates
- Full-Stack Observability



SOLUTION OVERVIEW

ProcureFlow at a Glance

5 + 1



Backend microservices + React frontend, fully containerized

5



Business domains: Identity, Procurement, Finance, Documents, AI Copilot

0



Database servers to manage — fully serverless data layer (DynamoDB + S3)

EKS



Deployed on Amazon EKS with GitOps continuous delivery via ArgoCD



SOLUTION OVERVIEW

Technology Stack



Frontend

React · TypeScript · Tailwind · Vite



Backend

Node.js · TypeScript · Express



Containerization

Docker



Orchestration

Amazon EKS



Infrastructure

Terraform



GitOps

ArgoCD · Helm



CI/CD

GitHub Actions



AI

Amazon Bedrock



Database

DynamoDB

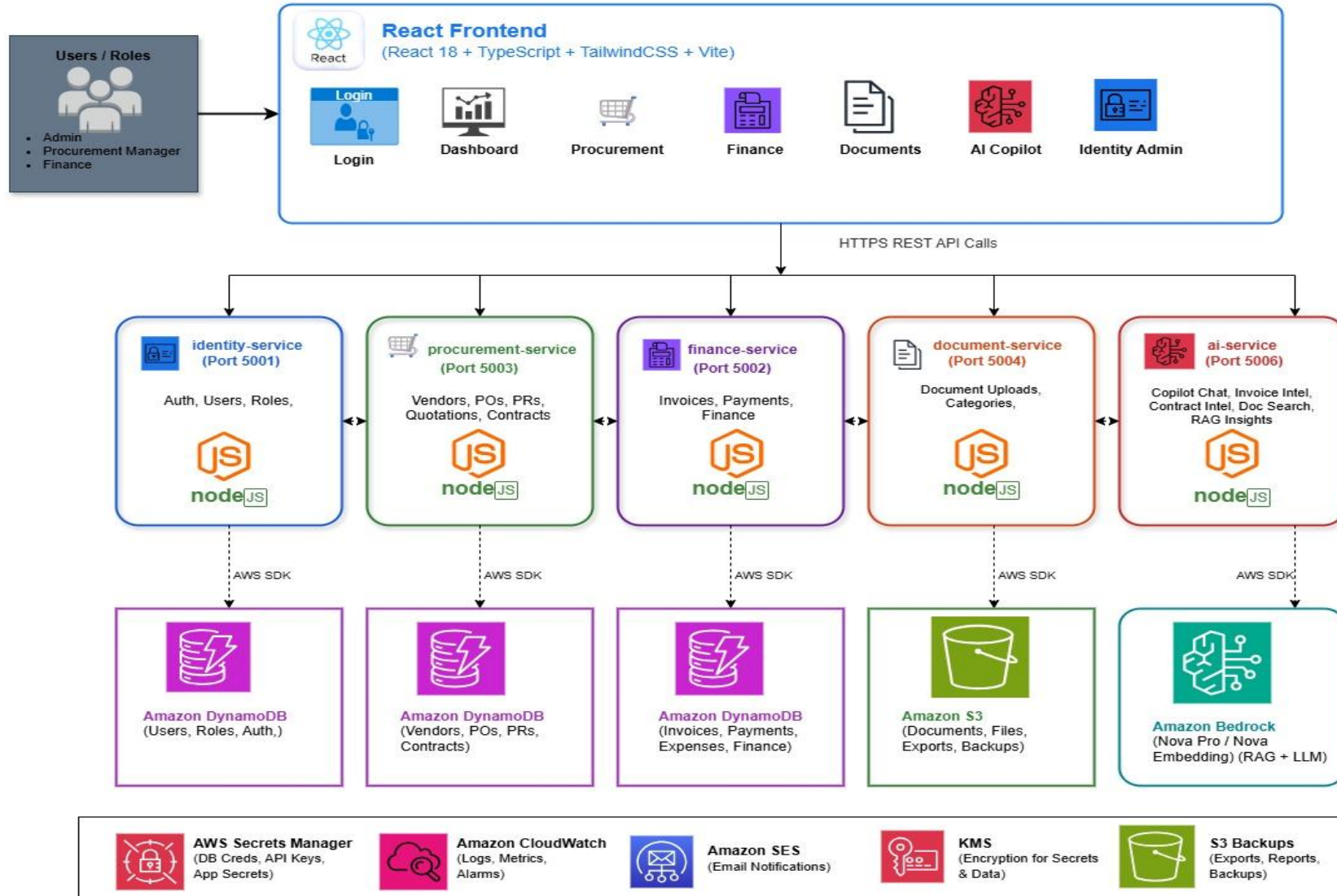


Storage

Amazon S3



ProcureFlow – Application Flow Architecture



Single Frontend · 5 Independent Microservices · Database-per-Service Pattern · AI Service Integrated with Amazon Bedrock



WALKTHROUGH · AI

AI-Powered Risk Intelligence

How a Risk Analysis Request Flows



Structured Risk Output

Every AI analysis returns a structured object the application can act on directly:

riskScore Numeric 0–100 confidence score

riskLevel low · medium · high · critical

findings

Duplicate invoice

Missing PO

Overbilling

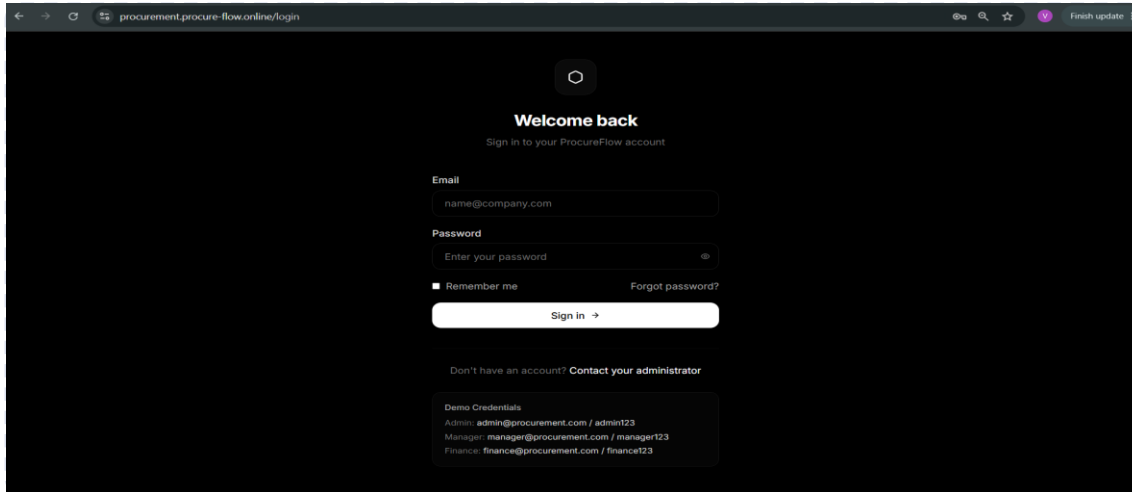
Contract violation

Role-scoped & IRSA-authenticated: a vendor user only ever sees their own records, and every Bedrock call runs with no hardcoded API keys.

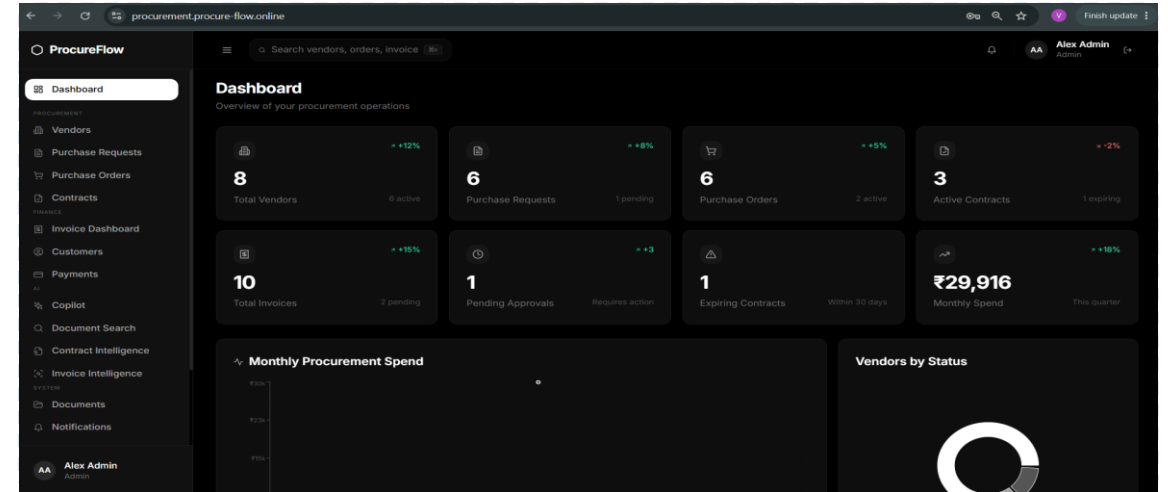


WALKTHROUGH · APPLICATION

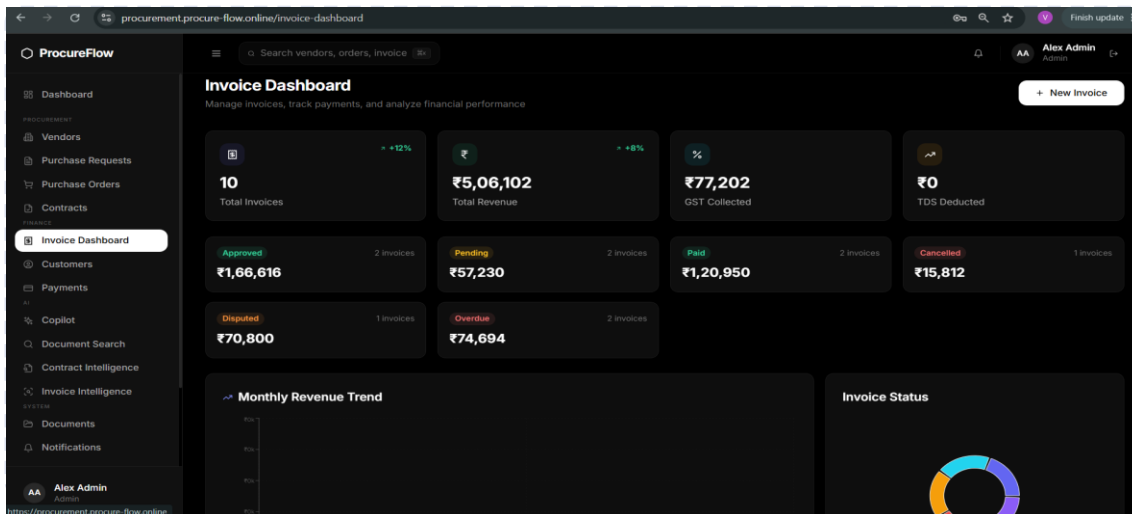
ProcureFlow in Action



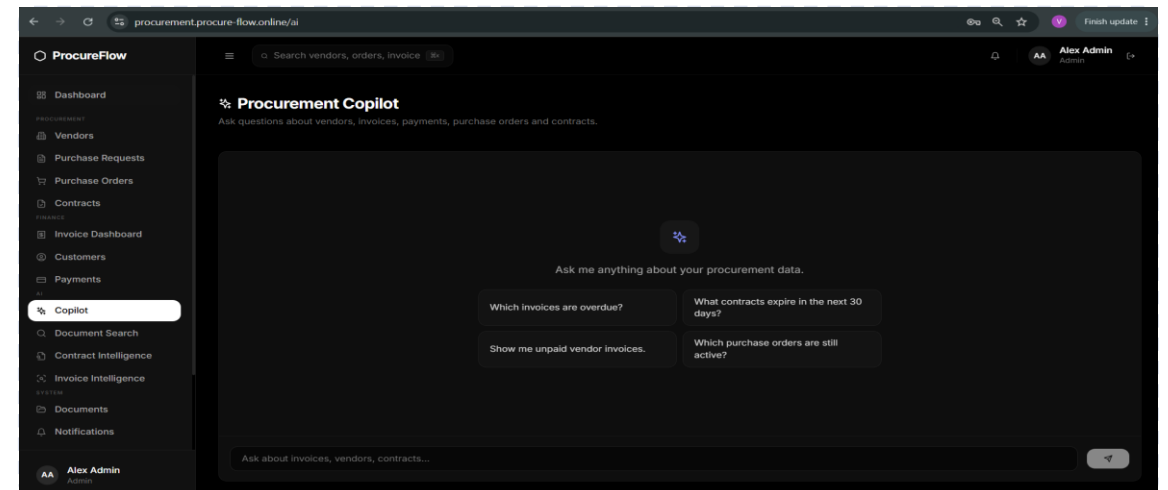
Login Page



Dashboard & Procurement



Finance — Invoices & Payments



AI Copilot — Risk & Contract Insights



WALKTHROUGH · APPLICATION

ProcureFlow in Action

ProcureFlow

Purchase Orders

Contracts

Invoice Dashboard

Customers

Payments

Copilot

Document Search

Contract Intelligence

Invoice Intelligence

Documents

Notifications

Reports

Audit Logs

Users

Settings

Alex Admin

Search vendors, orders, invoice

AA Alex Admin

Invoice Intelligence

Automated invoice validation — duplicate, mismatch, overbilling & contract-violation detection with an AI summary.

Search Invoices...

INV-INV7

Meridian Healthcare Systems · ₹1,55,760

INV-INV3

Global Tech Components · ₹32,450

INV-INV6

Brighorath Retail Group · ₹99,120

INV-INV1

Acme Industrial Supplies · ₹21,630

INV-INV2

Northwood Office Solutions · ₹10,855

INV-INV10

Vertex Packaging Ltd · ₹15,612

INV-INV9

Union Square Hospitality · ₹24,780

INV-INV5

Reveria Equipment Rentals · ₹70,800

INV-INV3

Global Tech Components · Due Jul 12, 2026

Re-analyze

62 mid

Medium Risk

1 finding(s) · rule engine v1

Findings

Missing Reference

Medium

Invoice lacks matching purchase order reference at submission time.

AI Summary

Automated risk analysis completed — score 62/100 (medium).

RECOMMENDATIONS

Route to finance lead for manual review.

Verify against original purchase order.

ProcureFlow

Purchase Orders

Contracts

Invoice Dashboard

Customers

Payments

Copilot

Document Search

Contract Intelligence

Invoice Intelligence

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Alex Admin

Search vendors, orders, invoice

AA Alex Admin

Contract Intelligence

Extract key terms, clauses and renewal risk from uploaded contract documents.

CONTRACT DOCUMENTS

acme-contract-2026.pdf

Jun 22, 2026

global-tech-contract-2026.pdf

Jun 22, 2026

acme-contract-2026.pdf

240 KB · application/pdf

Re-analyze

Vendor: Acme Industrial Supplies (VEN-1001) Contract No.: CTR-C1

Value: ₹4,80,000 Expiry: Apr 7, 2027

Auto-renewal requires 60 days notice to prevent, which may pose a moderate risk if not managed properly.

RENEWAL RISK

Medium

289 days to expiry

Summary

Annual industrial supply agreement with Acme Industrial Supplies for USD 480,000, effective from 2025-06-27 to 2027-04-07. Includes termination, liability, payment, and renewal terms.

Key Terms

TERMINATION: 30 days written notice

LIABILITY: Capped at contract value, no indirect or consequential damages

ProcureFlow

Purchase Orders

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AA Alex Admin

INV-INV3

Pending Vendor Invoice

Download PDF

PARTY

Global Tech Components

GSTIN: 29ABCDE1234F1Z5

DATES & REFERENCE

Issue Date: May 28, 2026

Due Date: Jul 12, 2026

PO Number: PO-PO4

LINE ITEMS

#	Description	HSN/SAC	Qty	Rate	Taxable	GST	Total
1	Seeded line item	---	1 unit	₹27,500	₹27,500	₹4,950	₹32,450

TAX BREAKDOWN

Sub Total (Taxable)	₹27,500	Gross Amount	₹32,450
CGST	₹2,475	Company Receivable	₹32,450
SGST	₹2,475	Vendor Payable (after TDS)	₹32,450
Total GST	₹4,950		

NOTES

Seeded document

ProcureFlow

Purchase Orders

Contracts

Invoice Dashboard

Customers

Payments

Copilot

Document Search

Contract Intelligence

Invoice Intelligence

Documents

Notifications

Reports

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Search vendors, orders, invoice

AA Alex Admin

Documents

Manage procurement documents and files

Upload Document

Vendor Certificate

All Categories

INV-INV7.pdf

vendor_certificate · 67.5 KB · 2h ago

acme-contract-2026.pdf

contract · 240.0 KB · 2d ago

po4-globaltech.pdf

purchase_order · 240.0 KB · 2d ago

sterling-invoice-inv4.pdf

invoice · 240.0 KB · 2d ago

globaltech-iso-cert.pdf

vendor_certificate · 240.0 KB · 2d ago

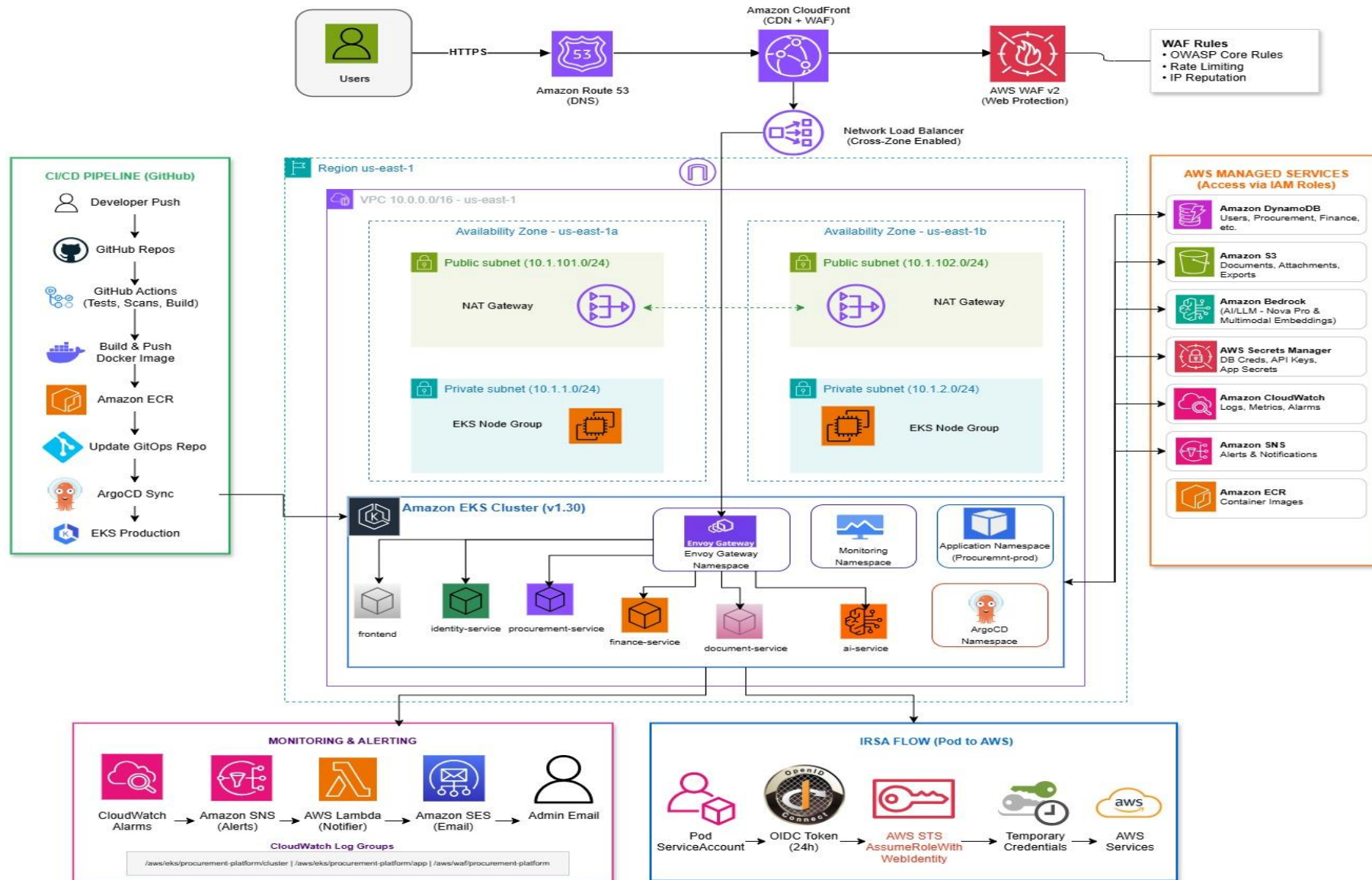
global-tech-contract-2026.pdf

contract · 240.0 KB · 2d ago



Procurement Platform – Cloud Infrastructure Architecture

AWS – Production Environment





WALKTHROUGH · CLOUD

How Traffic & Identity Flow Through AWS

Request Path



Route 53

DNS resolution for the platform domain



CloudFront + WAF v2

CDN edge with OWASP core rules, rate limiting, IP reputation filtering



Network Load Balancer

Cross-zone enabled, hands off to the in-cluster Envoy Gateway



EKS 1.30 across 2 AZs

Managed node groups in us-east-1a and us-east-1b

Single AWS account, us-east-1, entirely Infrastructure-as-Code — no manually created resources.

IRSA Flow — Pod to AWS

Pod ServiceAccount

OIDC Token (24h)

AWS STS — AssumeRoleWithWebIdentity

Temporary Credentials

AWS Services

Every pod authenticates with a short-lived, auto-rotated token — there is no long-lived AWS key anywhere in the cluster.

Monitoring & alerting hangs off the same VPC: CloudWatch Alarms → SNS → Lambda → SES, with dedicated log groups per EKS cluster, app, and WAF.



WALKTHROUGH · CLOUD

Network Architecture & Isolation

VPC 10.0.0.0/16 · us-east-1

Availability Zone — us-east-1a

Public Subnet



NAT Gateway

Private Subnet



EKS Node Group

Availability Zone — us-east-1b

Public Subnet



NAT Gateway

Private Subnet



EKS Node Group

- Custom VPC, 2 AZs, public + private subnet pair per AZ
- Private subnets host all EKS worker nodes — no direct internet exposure
- Prod: one NAT Gateway per AZ (high availability). Dev: single shared NAT Gateway (cost optimization) — a deliberate, environment-aware tradeoff
- VPC CNI with prefix delegation raises pod density per node from ~17 to ~110, letting 2 nodes host the entire platform + ArgoCD + monitoring stack
- Bastion host available for break-glass access — never required for routine operations



WALKTHROUGH · IAC

Everything Defined as Code

17 Reusable Terraform Modules



Remote state

S3 backend + DynamoDB state locking — safe for concurrent CI/local runs



Workspaces

Terraform Workspaces (dev/prod) — same code, environment-specific .tfvars



prevent_destroy

Lifecycle protection on ACM, Route 53, ECR — irreplaceable resources can't be accidentally destroyed



100% tagged

Environment, Owner, Project, ManagedBy applied via provider default_tags

17 Reusable Modules

Single Source of Truth

Terraform Workspaces

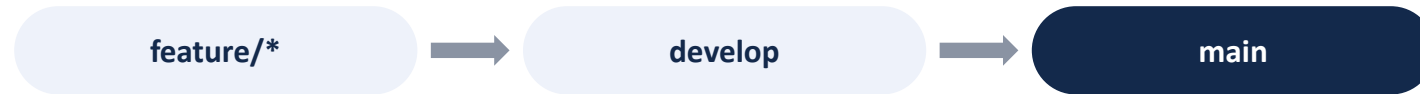
Automated Infrastructure



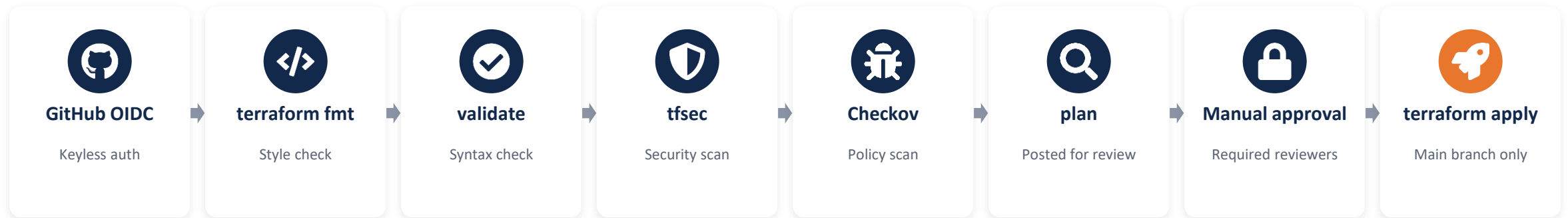
WALKTHROUGH · IAC

Infrastructure Pipeline & Governance

Branching Strategy — mirrored across infra / gitops / app repos



Terraform CI/CD Pipeline

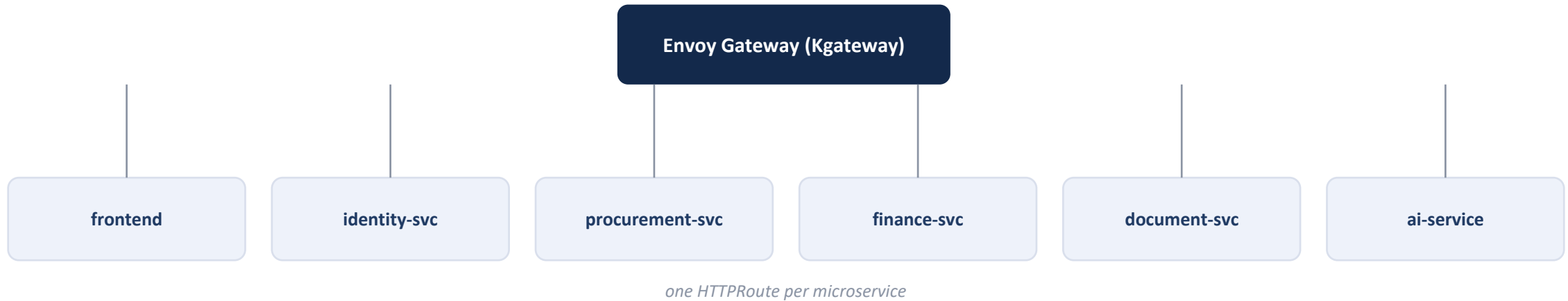


- terraform apply runs ONLY after merge to main AND a manual approval gate (GitHub Environment with required reviewers)
- Cost optimization: nightly targeted destroy of the compute layer (VPC/EKS) while the data layer and singletons (ECR/ACM/Route 53) stay protected and untouched



WALKTHROUGH · DEPLOYMENT

Running on Amazon EKS

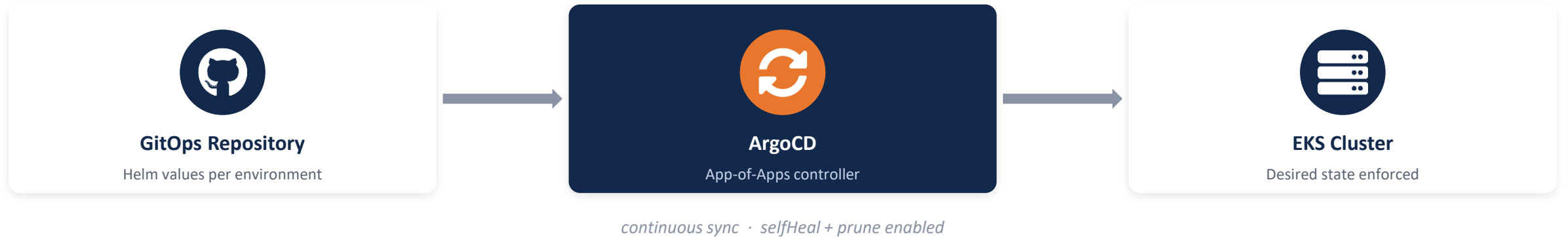


- EKS 1.30, managed node groups, dedicated procurement-prod namespace — isolated from kube-system / argocd / monitoring
- Routing via Kgateway, an Envoy-based implementation of the Kubernetes Gateway API — the modern successor to nginx Ingress
- One HTTPRoute per microservice, all attached to a single shared Gateway per environment
- The Gateway's Network Load Balancer is internet-facing and TLS-capable, fronting all 6 services through one entry point



WALKTHROUGH · DEPLOYMENT

GitOps: Git as the Source of Truth

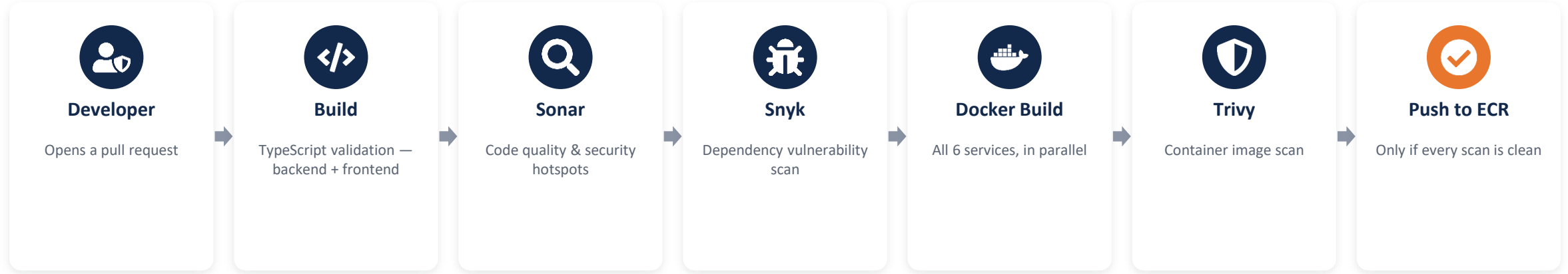


- ArgoCD continuously syncs desired state from a dedicated GitOps repository
- App-of-Apps pattern: one Application per environment (dev / prod), each pointing to environment-specific Helm values
- selfHeal + prune enabled — any manual cluster drift is automatically reverted to match Git
- Image tags and IRSA role ARNs are injected automatically by the deploy pipeline — no manual YAML editing, ever



WALKTHROUGH · CI/CD

Continuous Integration — Shift-Left Security

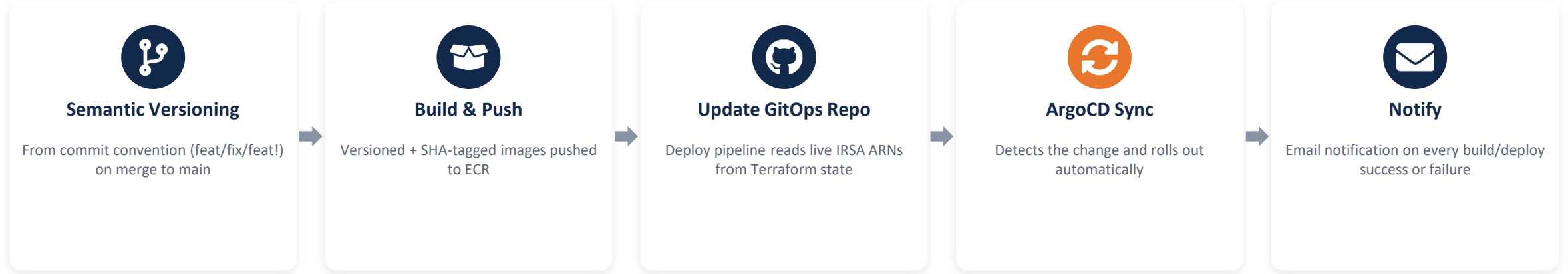


- Every pull request receives automated scan results before a merge is allowed
- Security scanning happens at code, dependency, and container layers — before anything reaches the registry



WALKTHROUGH · CI/CD

Continuous Deployment to Production

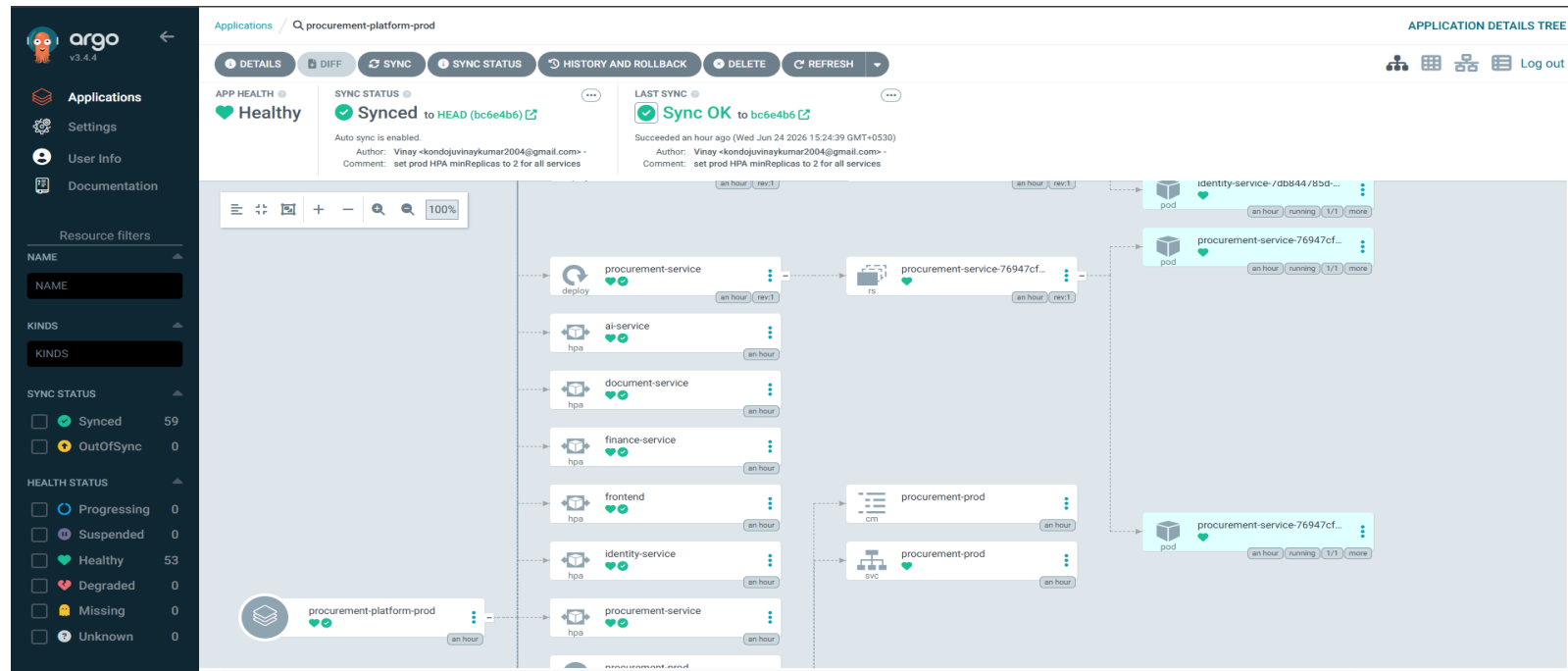
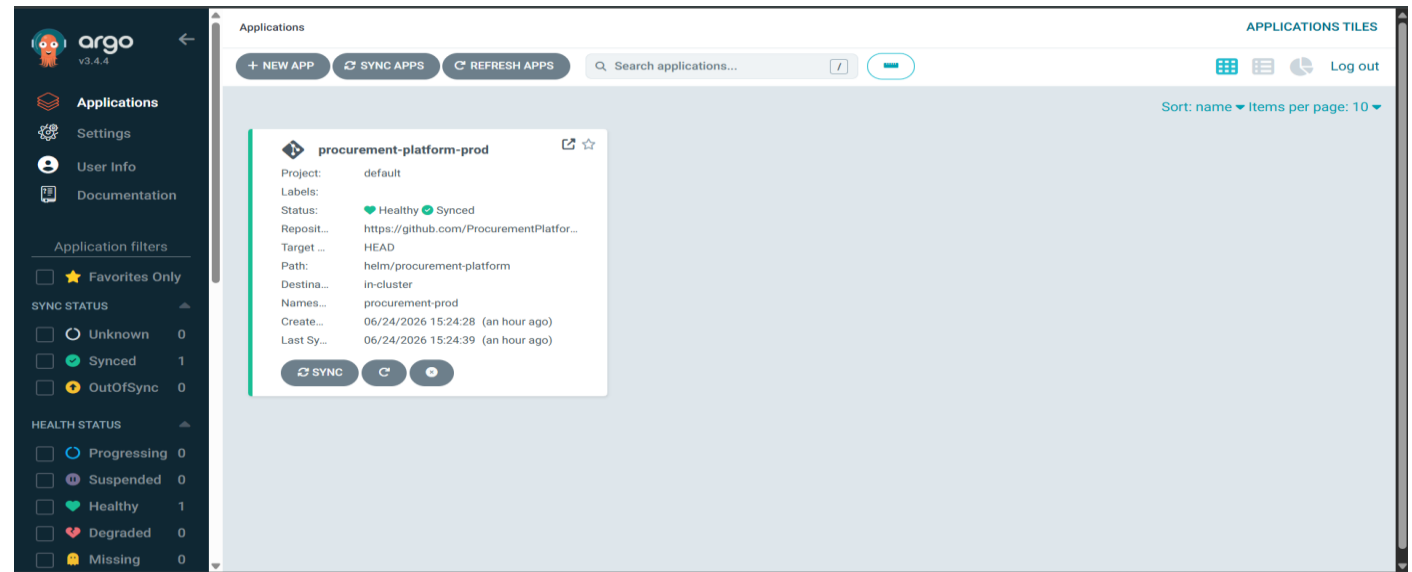


- Zero manual kubectl commands anywhere in the deploy path
- The same pipeline pattern runs identically for dev and prod, parameterized entirely through GitOps repo branches



WALKTHROUGH · DEPLOYMENT

ArgoCD in Action



ArgoCD Dashboard — Application Sync Status



WALKTHROUGH · CLOUD SERVICES

Managed AWS Services Powering the Platform



DynamoDB

15 tables, serverless, pay-per-request —
all application data



S3

Documents, presigned URLs, versioned +
KMS-encrypted



Secrets Manager

IRSA-authenticated, zero hardcoded
secrets



KMS

Customer-managed keys for DynamoDB,
S3, Secrets Manager



SES + SNS + Lambda

Automated alerting pipeline



Route 53 + ACM

DNS resolution and TLS certificate
management



CloudFront + WAF

Built and verified, ready to activate at the
edge



WALKTHROUGH · OPERATIONS

Full-Stack Observability

Cluster & Application Monitoring



Prometheus + Grafana + Alertmanager

kube-prometheus-stack, deployed via the same GitOps-managed cluster



Custom application metrics

Request count & latency histograms instrumented directly into every backend service



Auto-provisioned dashboards

Custom Grafana dashboard auto-loads on every cluster rebuild — zero manual setup



Slack alerting

Alertmanager routes cluster/application alerts directly to Slack

AWS-Level Alert Pipeline



CloudWatch Alarms



Amazon SNS



AWS Lambda



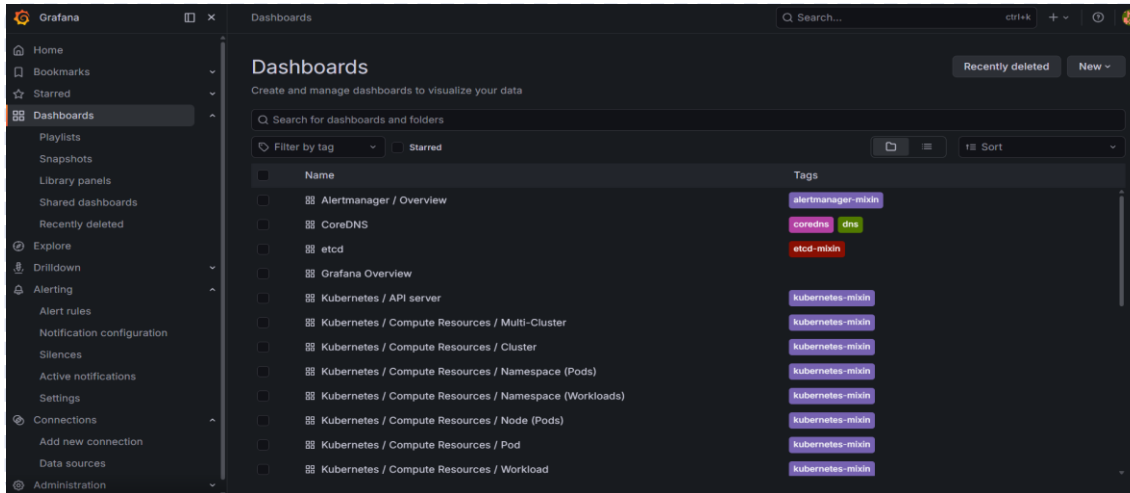
Amazon SES

Formatted HTML email delivered to the admin inbox on every alarm.



WALKTHROUGH · OPERATIONS

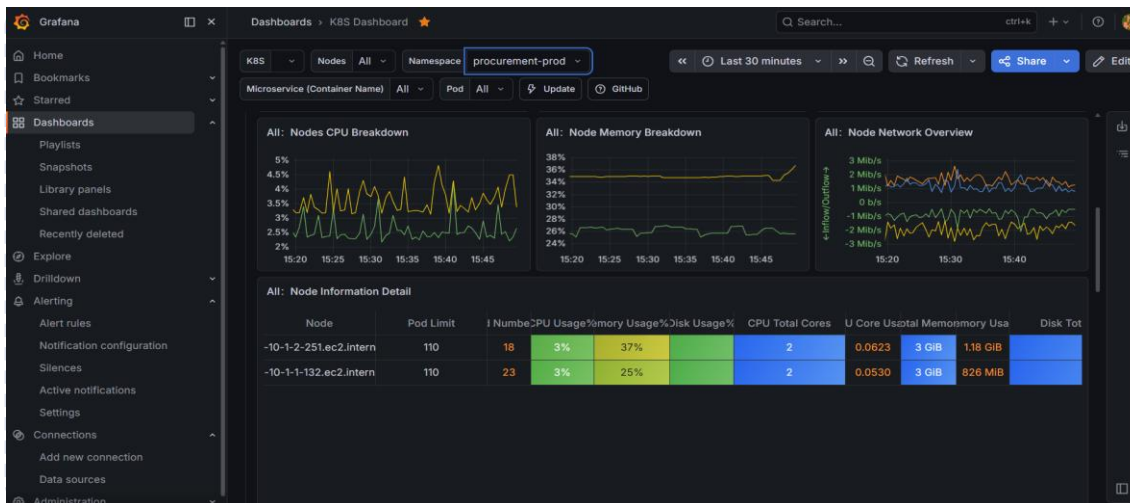
Monitoring Dashboards in Action



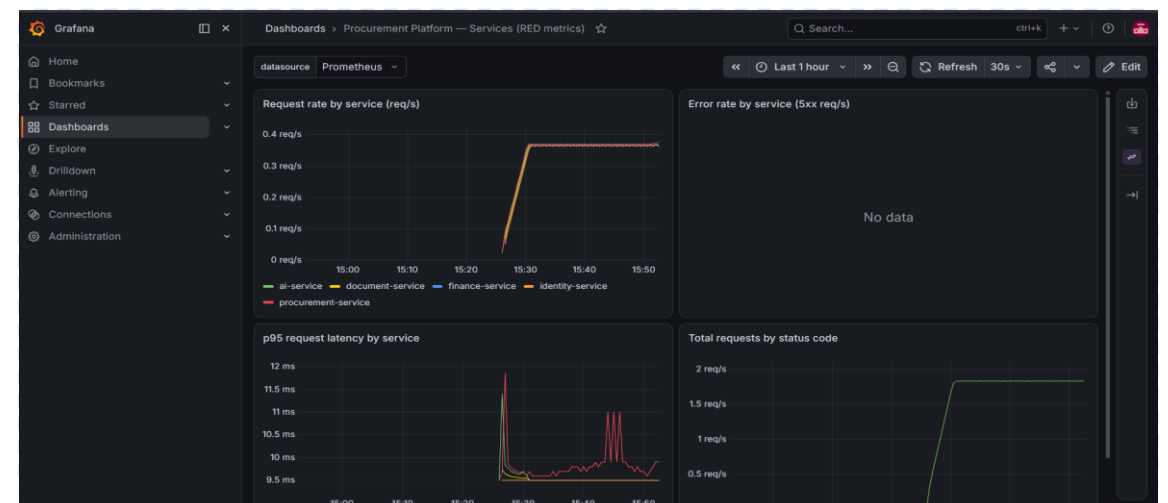
Grafana — Dashboard



Kubernetes Dashboard



Grafana — Service Latency & Requests



Procurement-Platform Request Rate Dashboard



WALKTHROUGH · SECURITY

Defense in Depth



Default-deny NetworkPolicies

Every pod blocked by default — explicit allow rules only for the Gateway and known service-to-service calls



Zero Kubernetes Secret objects

Application secrets fetched directly from AWS Secrets Manager at runtime via IRSA — never stored in the cluster



Hardened containers

Non-root containers, dropped Linux capabilities, multi-stage Docker builds with a separate prod-only dependency stage



GitHub OIDC everywhere

All CI/CD AWS access via GitHub OIDC — no static access keys anywhere in any repository



WALKTHROUGH · SCALABILITY

Built to Scale and Stay Up

2 → 10

Replicas per service via Horizontal Pod Autoscaler

80%

CPU utilization threshold that triggers scale-out

2 AZs

Multi-AZ EKS node groups, no single point of failure

0

Single points of failure — per-AZ NAT Gateways in production

- PodDisruptionBudgets ensure minimum availability during node maintenance and upgrades
- Multi-AZ EKS node groups together with per-AZ NAT Gateways in production eliminate single points of failure
- ArgoCD's ignoreDifferences tuning ensures the GitOps controller and the autoscaler never fight each other over replica count



SUMMARY

Security Posture — End to End



Zero static AWS credentials anywhere

GitHub OIDC + IRSA exclusively — no access keys in any repository or pod



Encrypted at rest and in transit

KMS customer-managed keys for data at rest; TLS for everything in motion



Automated scanning at every layer

Code (SonarCloud), dependencies (Snyk), containers (Trivy), infrastructure (tfsec)



Manual human approval gate

Required before any production infrastructure or deployment change



Default-deny network posture

Principle of least privilege enforced on every IAM and IRSA role

Thank You



Live Application

procurement.procure-flow.online



GitHub Organization

[ProcurementPlatform](#)



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